Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions**

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Dataset at a Glance

3,900

Purchases

18

Features

$59.76

Avg. Spend

3.75

Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings, Shipping

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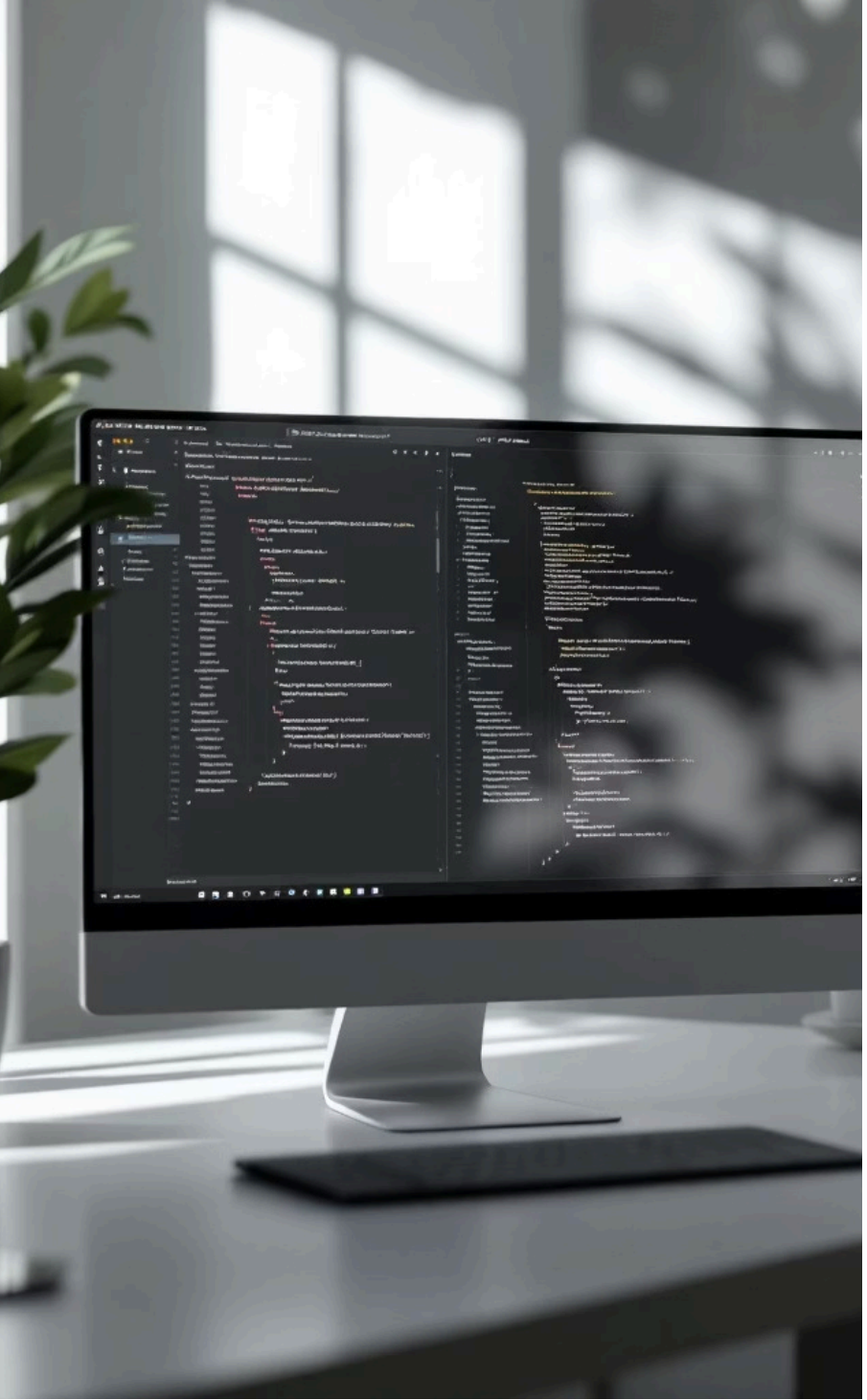

Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean & Standardize

Imputed missing ratings; renamed columns to snake_case

03

Feature Engineering

Created age_group and purchase_frequency_days

04

Load to PostgreSQL

Cleaned DataFrame exported for SQL analysis

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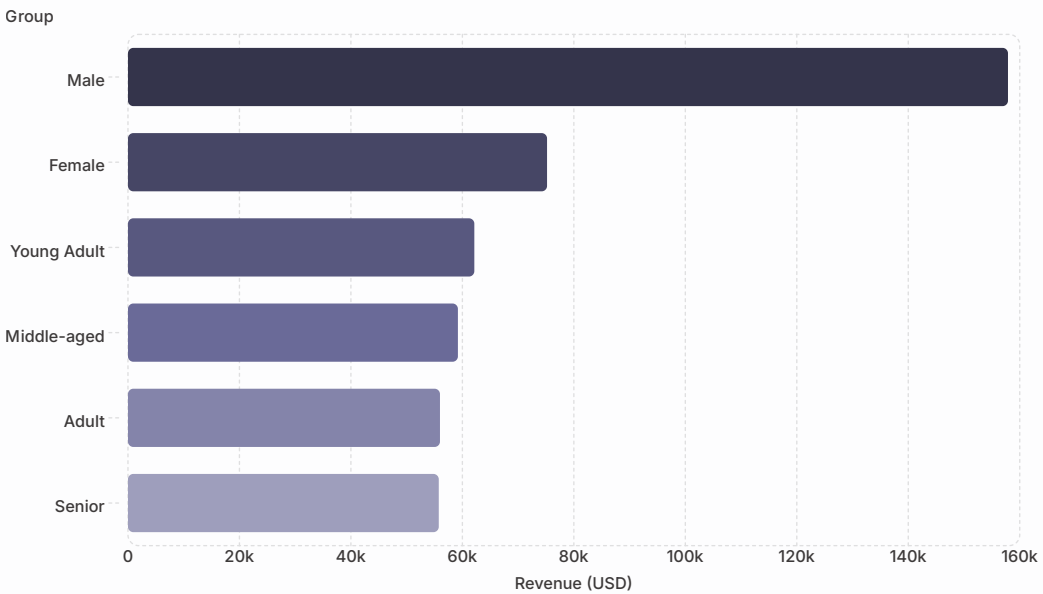
Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults are the **top revenue age group**
- Express shipping users spend slightly more ($60.48 vs. $58.46 Standard)
- 839 discount users still spent **above average**

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Top-Rated Products

Top 5 by Avg. Review Rating

1	Gloves — 3.86
2	Sandals — 3.84
3	Boots — 3.82
4	Hat — 3.80
5	Skirt — 3.78

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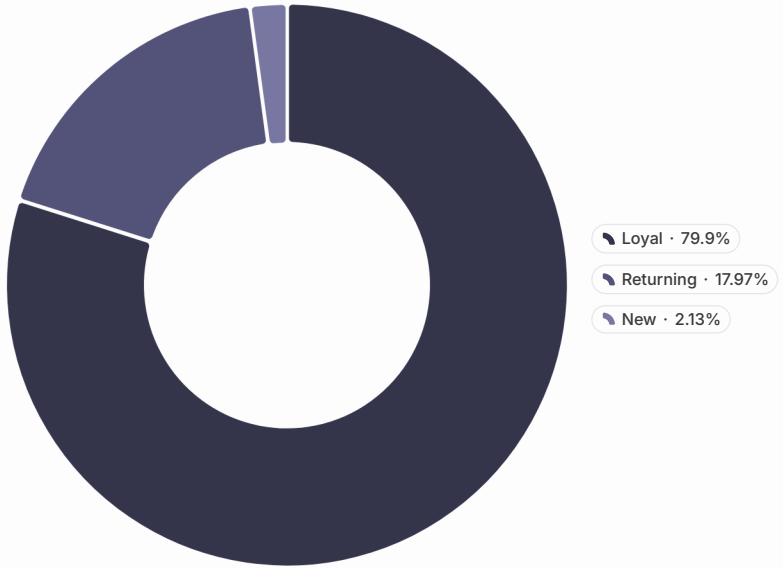
Customer Segmentation



Segment Breakdown

Loyal

3,116 customers

Returning

701 customers

New

83 customers

Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 non-subscribed

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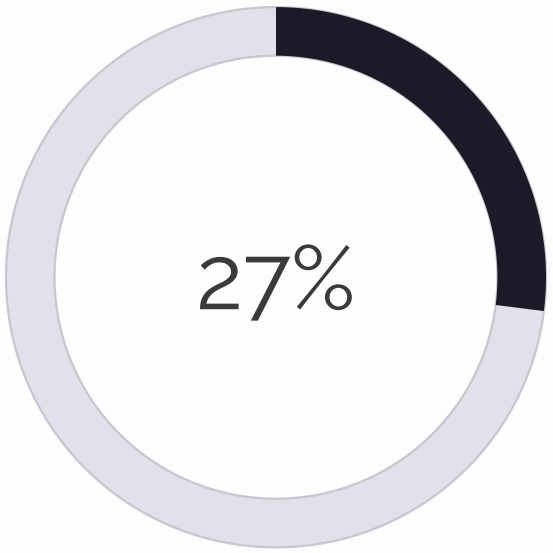
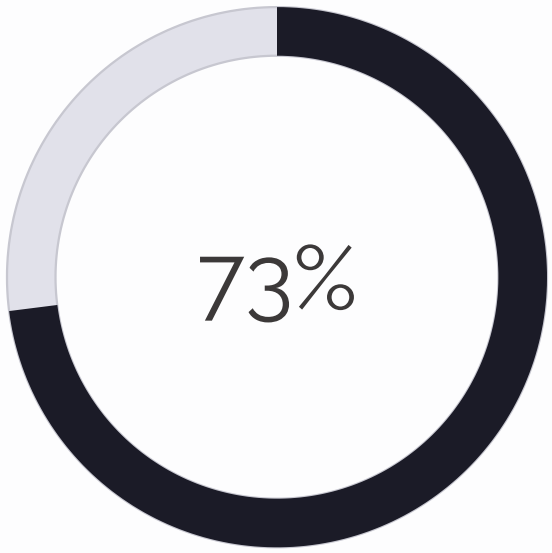
Subscriptions & Discounts



Subscribers

1,053 customers · $62,645 total revenue



Non-Subscribers

2,847 customers · $170,436 total revenue

Most Discount-Dependent Products

1. Hat 50%
2. Sneakers 49.66%
3. Coat 49.07%
4. Sweater 48.17%
5. Pants 47.37%

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Top Products by Category

Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

Outerwear

- Jacket — 163
- Coat — 161

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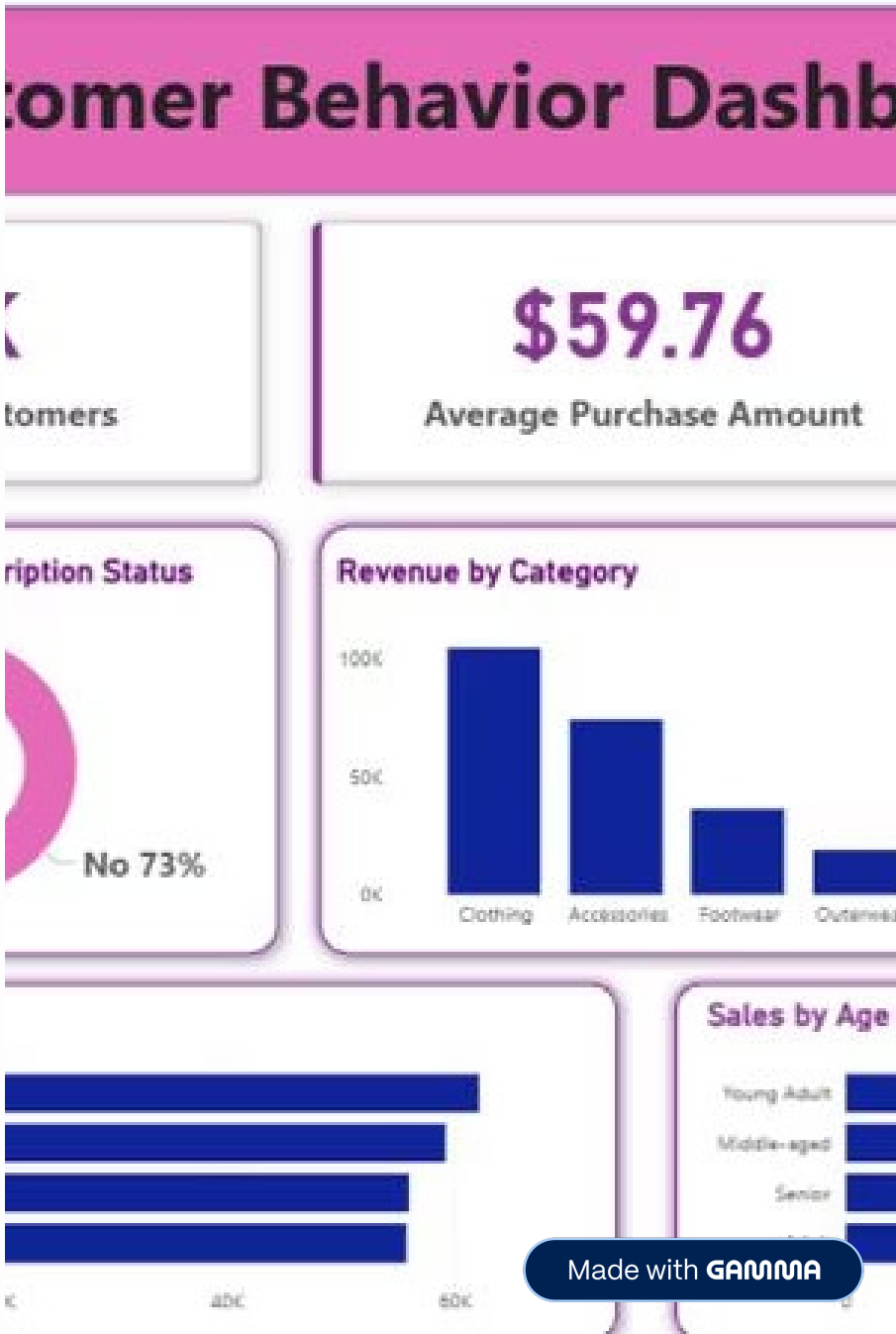
Power BI Dashboard

Interactive dashboard with filters for Subscription, Gender, Category & Shipping — surfacing KPIs and visual trends across all segments.



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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Product Positioning

Highlight top-rated & best-selling items in campaigns

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